Question			Answer	Marks	AO	Guidance	
12	(a)		$s = ut + \frac{1}{2}at^2$ with $s = -1, u = 0, t = 1.1$	M1	1.1a	Allow sign errors	
			$a = -1.65$ ms^{-2}	A1	1.1	Must be 3 sf.	Allow $a = 1.65$
				[2]			
12	(b)		$x = 25\cos 40° t$	B1	1.1	Oe eg $t = \frac{x}{25\cos 40°}$	$t = 0.0522x$ or $\frac{x}{19.15}$ or $\frac{x}{19.2}$
			$y = 25\sin 40° t + \frac{1}{2}at^2$	M1	3.3	Forms equation of motion in the vertical direction	Soi. Allow sign errors. Do not allow if $a = \pm 9.8$ used here or subsequently
			$y = 25\sin 40°\left(\frac{x}{25\cos 40°}\right) + \frac{1}{2}a\left(\frac{x}{25\cos 40°}\right)^2$	M1	2.1	Substitutes expression for t in their y equation	
			$y = 0.839x - 0.00225x^2$	A1	1.1	Cao. Coefficients must be 3sf.	
				[4]			
12	(c)		When $x = 40, y = 30.0[> 15]$	M1	3.4	Use their model and $x = 40$ (29.72 if exact values used)	
			So the ball goes over the crater's edge	A1	3.2a	Conclusion based on correct working. No FT from wrong (b)	Possible interpretation of the question gives $x = \sqrt{40^2 - 15^2} = 5\sqrt{55}$ $= 37.1$ giving $y = 28.0$
			Alternative method				
			When $y = 15, x =$18.8, 354	M1		Use the model with $y = 15$	
			So the ball is above 15m as 40 m is between these values	A1		Conclusion based on correct working No FT for wrong (b)	Possible interpretation $x = \sqrt{1375} = 37.1$ which is between these values.
				[2]		When $x = 40, t = 2.09$ when $y = 15$, $t = 0.983, 18.5$	